



The Influence of Strategic Prospectivity, Absorptive Capacity, and Boundary-Spanning Search on Exaptation

Chaoyong Tang¹ · Yongzhi Shi² · Ruili Sun¹

Received: 13 August 2022 / Accepted: 25 September 2024 / Published online: 4 November 2024
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024

Abstract

This study forms a moderated mediating model by introducing the two variables of absorptive capacity and boundary-spanning search to explore the influence of strategic prospectivity on exaptation. The paper focuses on the mediating role of boundary-spanning search and the moderating role of absorptive capacity. The results show that (1) strategic prospectivity has a significant positive impact on exaptation and boundary-spanning search mediates the relationship between strategic prospectivity and exaptation. (2) Potential absorptive capacity and realized absorptive capacity strengthen the relationship between boundary-spanning search and exaptation. (3) Potential absorptive capacity regulates the mediating role of boundary-spanning search. The hypothesis that put forward that realized absorptive capacity positively regulates the mediating effect of boundary-spanning search has not been verified. The research findings enrich our understanding of the impact mechanisms of exaptation and offer valuable insights for Chinese enterprises seeking to promote exaptation practices.

Keywords Strategic prospectivity · Exaptation · Boundary-spanning search · Absorptive capacity

Introduction

Masks do not need to be removed for meals; mobile phones that integrate the functions of communication, photography, photo editing, and bank cards; drones that expand their application from the military to civilian fields; and webcast systems

✉ Chaoyong Tang
tcyky2012@163.com

¹ School of Economics and Management, Hebei Agricultural University, No. 2596, Lekainan Street, Lianchi District, Baoding City 071001, Hebei Province, China

² Modern Educational Technology Center, Hebei Agricultural University, Baoding City 071001, Hebei Province, China

used for multi-purpose applications all involve a new research category, namely, exaptation (Ren et al., 2019). As a reverse innovation mechanism that uses the answer to find the problem, exaptation has gradually become an important source of innovation. Exaptation has also made a significant contribution to innovation and can even cause major changes to an industry or even open up a whole new industry (Dew et al., 2004). However, exaptation is not the result of the survival of the fittest approach (Villani et al., 2011). Rather, exaptation is a process with unexpected characteristics, which can only happen if a forward-looking plan that can actively foresee the future is adopted to play a creative role and establish new development opportunities (Ren et al., 2019). In order to better activate and promote exaptation, enterprises can use strategic prospectivity to improve the probability of exaptation occurring. As a unique attribute of “prescient” organizations (Miles et al., 1978), strategic prospectivity attaches great importance to capturing new knowledge, new technology and new markets from outside the enterprise. In addition, exaptation actively changes an organization’s ideological cognition, reengineers the organizational structure and innovates the organizational culture. These functions serve to dynamically adjust the product and market positioning according to the market opportunities, thus realizing the coupling between the existing technology system and new fields. Finally, exaptation enhances the existing knowledge and technology reconstruction abilities, which is of great value for adaptation (Xu & Yu, 2020).

Strategic prospectivity and exaptation are by no means in a simple direct linear relationship. Some mediating mechanism may also exist, but previous studies have paid little attention to this. As an important variable in constructing the “second-order capability” of an organization (Rosenkopf & Nerkar, 2001), boundary-spanning search can provide rich knowledge resources for enterprises’ exaptation. It can also provide new application fields and marketing opportunities for existing technologies and products and improve the possibility of exaptation. Moreover, the novel, emerging, and first-of-its-kind knowledge acquired through boundary-spanning search can cultivate open-mindedness and organizational creativity and improve the ability to deconstruct and reconstruct existing technologies (Dew et al., 2004). Importantly, this new knowledge also contributes to the ability to tap into new functionalities of existing technologies, thereby facilitating the next generation of exaptation. In addition, some studies have found that corporate boundary-spanning search intention is inseparable from the support of strategic prospectivity, which has an obvious positive impact on boundary-spanning search (Sidhu et al., 2004). Therefore, boundary-spanning search may be a new path along which strategic prospectivity can drive exaptation. However, few studies thus far have introduced boundary-spanning search into theoretical models to analyze its inherent logic in the relationship between strategic prospectivity and exaptation.

A link mechanism does exist between strategic prospectivity and exaptation. However, the effect of boundary-spanning search depends on whether the enterprise has the ability to effectively identify, transform and apply external heterogeneous information to innovate the existing knowledge system. This ability is mainly reflected in the absorptive capacity of the enterprise (Cohen & Levinthal, 1990). Existing research shows that absorptive capacity can improve the sensitivity and responsiveness to perceived changes in the external environment and market

opportunities. Absorptive capacity also helps enterprises to judge future technological changes, get rid of path dependence and organizational rigidity, and internalize, reconstruct, and allocate the innovative resources obtained by boundary-spanning search that are required to promote the exaptation process (Kastelli et al., 2022). Other studies have found that the intensity of absorptive capacity affects the impact of boundary-spanning search on exaptation, through the effective identification and transformation of knowledge (Zahra & George, 2002). In addition, absorptive capacity may also accurately grasp current market and technological trends, which in turn have an impact on the mediating effect of boundary-spanning search (Lev et al., 2009). Therefore, in addition to the moderating effect of absorptive capacity between boundary-spanning search and exaptation, absorptive capacity may also strengthen the intermediary mechanism of boundary-spanning search between strategic prospectivity and exaptation. However, very little existing literature discusses this topic, which is not conducive to deepening understanding of the mechanism through which strategic prospectivity influences exaptation.

The academic value of this paper is mainly reflected in the following: (1) introducing strategic prospectivity into the context of exaptation and exploring the mechanism of strategic prospectivity on exaptation. These contributions enrich and expand existing research. (2) Based on boundary-spanning search theory, this paper investigates the intermediary role of boundary-spanning search between strategic prospectivity and exaptation. This contribution helps to open the black box of strategic prospectivity-driven exaptation. (3) This paper also integrates absorptive capacity into the theoretical model of strategic prospectivity, boundary-spanning search and exaptation. The regulation mechanism of absorptive capacity is also discussed, and the boundary conditions in which strategic prospectivity indirectly affects exaptation are further revealed.

Theories

Exaptation first originated from evolutionary biology. Mokyr (2000) initially introduced the concept into management, and interpreted exaptation as “the success of a certain technology after being accidentally used in another field.” Subsequently, Gregory (2008) espoused that exaptation refers to existing components, which initially had no (or additional) functions. Exaptation refers to the components being rearranged into a new and more complex form, with new functions. Arthur (2007) understood exaptation as follows: when a new environment emerges, or when a new requirement arises in an application area, old technology or old basic principles are used and transformed by “extending” them to adapt to the new environment. Ganzaroli & Pilotti (2011) also pointed out that exaptation means using the available existing functions to perform a new function, or selecting an environment to stimulate their potential. Existing studies have shown that, as a situational innovation method, exaptation may lead to the creation of new technological trajectories, markets or industries (Andriani & Carignani, 2014). As such, exaptation has an important driving role in enterprises’ technological innovation. Therefore, how to activate and promote exaptation has become an important issue for scholars to explore. From

the perspective of mergers and acquisitions (M&A), Dew et al. (2004) found that a certain technology of an enterprise may be re-stimulated for new uses after M&A. From the perspective of user innovation, Andriani et al. (2017) pointed out that users can modify existing products according to their subjective judgments, or redefine how products are used and discover their new functions. Ganzaroli & Pilotti (2011) further pointed out that combining existing knowledge capabilities with new fields and creating an interaction between producers and users can help improve the probability of successful exaptation. Dew et al. (2004), Garud et al. (2016), and Abatecola et al. (2016) emphasized the importance of knowledge reserve and knowledge diversity in enhancing exaptation, from the perspective of knowledge management. From the perspective of corporate culture, Ganzaroli et al. (2014) should that innovative culture can make companies break away from institutional constraints. Such a culture can also enhance companies' ability to change, promote the interaction between cognition of research and development (R&D) personnel and organizational resources, and stimulate new innovation methods. Andriani & Carignani (2014) revealed that technical complexity is a necessary condition for scaling up exaptation from a technical complexity perspective. As can be seen, previous research on the formation mechanism of exaptation was either limited to the external perspective of the enterprise (M&A, user innovation) or focused on the internal perspective of the enterprise (corporate culture, knowledge management). However, these studies lacked the comprehensive effect of considering the internal and external factors of the enterprise together.

According to Miles et al. (1978), there are four kinds of strategic actors in an enterprise: the forerunner, the prophet, the latecomer and the unconscious. Strategic prospectivity is the most important feature of a foresight-type enterprise. An enterprise having foresight means that, in the face of a complex, ever-changing and highly competitive external environment, such enterprises pay more attention to searching for new technologies, new products and new markets. These enterprises tend to search for, identify and apply new market opportunities, and they carry out timely product innovation and market positioning. These steps enable the enterprise to achieve synergistic matching of technological innovation and management innovation, and to demonstrate the flexibility of organizational duality, so as to obtain long-term competitive advantages (Yang & Yu, 2016). This study found that strategic prospectivity pays attention to new opportunities in a changing market environment, tracks cutting-edge technologies, and acquires diversified original technologies to develop new products and develop new markets. In addition, strategic prospectivity also has the ability to predict and make strategic decisions about industry development. Meanwhile, strategic prospectivity also has a complete organizational system to coordinate resource allocation between departments, and to quickly restructure strategic decisions (Xu & Yu, 2020). Early research mainly focused on how to use strategic prospectivity to enhance corporate performance (Liu & Chen, 2007), thus revealing a new path for corporate performance improvement. Recent studies have found that a positive correlation exists between strategic prospectivity and innovation performance (Wei et al., 2015; Xu & Yu 2020). As can be seen, a close relationship exists between strategic prospectivity and corporate innovation. However, as

an important source of innovation, is exaptation also affected by strategic prospectivity, and what is the underlying logic of this impact? Theorists have yet to answer this question.

Boundary-spanning search is a tool and process used to obtain problem-solving solutions with cross-organizational learning as the main means. Boundary-spanning search is defined as a process in which enterprises acquire heterogeneous knowledge (such as external new ideas, new management and new technologies) by crossing cognitive, physical, and technological boundaries. The process also includes enterprises enhancing their organizational change ability, innovation ability, and adaptability through organizational learning and then continuously improving organizational innovation performance (Cyert & March, 1963; Katila & Ahuja, 2002). Previous studies have shown that boundary-spanning not only supports enterprises in their efforts to carry out product innovation, process innovation and market innovation, but also improves business activities, management level, and strategic decision-making modes. This is achieved by acquiring valuable diversified and heterogeneous knowledge resources in different fields (Pryor & Henley, 2017). Boundary-spanning also helps to enhance knowledge creation ability and technology research and development ability and promotes the technological innovation of enterprises (Ahuja & Lampert, 2001). In addition, studies have found that factors such as environmental dynamics, resource redundancy, and organizational forward-looking are key factors that drive organizations to conduct boundary-spanning search (Sidhu et al., 2004). However, existing literature is relatively lacking in the logic required to interpret the ins and outs of boundary-spanning search from a strategic prospectivity perspective; further research is needed.

Cohen & Levinthal (1990: 128) defined absorptive capacity as “the ability of an enterprise to identify the value of new knowledge from the external environment, digest it and finally apply it for commercialization purposes.” Previous studies have shown that absorptive capacity can directly and indirectly promote innovation performance (Chen & Miao, 2014; Cohen & Levinthal, 1990). In addition, absorptive capacity can also positively regulate the positive relationship between knowledge search, knowledge spillover, user-supplier knowledge acquisition, and innovation performance (Rothaermel & Alexandre, 2009; Tsai, 2001). In fact, when enterprises use boundary-spanning search to obtain external resources, strong absorptive capacity not only helps enterprises improve the organization’s sensitivity, resilience, and responsiveness to opportunities or threats; such enterprises can also quickly absorb the complementary resources needed to achieve the efficient integration, transformation, and application of new knowledge, new technologies, and new markets (Chen & Miao, 2014). In addition, according to the research of Zahra & George (2002), absorptive capacity includes both potential absorptive capacity and realized absorptive capacity. The former focuses on the identification and digestion of knowledge, while the latter focuses on the transformation and application of knowledge. This paper argues that both potential absorptive capacity and realized absorptive capacity affect the relationship between boundary-spanning search and exaptation through different mechanisms. Therefore, absorptive capacity (potential/realized absorptive capacity) is incorporated into the theoretical model, to further explore the boundary conditions in which strategic prospectivity affects exaptation.

The above research has achieved certain theoretical results, but some problems are still worthy of further discussion. (1) Previous research mostly focused on exploring the impact mechanism and formation path of exaptation from the perspectives of user innovation, mergers and acquisitions, and corporate culture. Theoretical discussion on the relationship between strategic prospectivity and exaptation has been lacking. (2) Previous studies have explored the intermediary mechanism of boundary-spanning search in the relationship between strategic prospectivity and innovation performance. However, the intermediary mechanism of boundary-spanning between strategic prospectivity and exaptation needs to be revealed. (3) Previous studies have paid little attention to the regulatory mechanism between boundary-spanning search and exaptation. This paper analyzes the nature and changes in the strength of the relationship between boundary-spanning search and exaptation under the condition of absorptive capacity. This approach will help make up for the lack of a regulatory mechanism between boundary-spanning search and exaptation. Based on the above, in order to make up for the shortcomings of existing research, this paper constructs a theoretical model of the relationship between strategic prospectivity and exaptation. Then, the impact mechanism of strategic prospectivity on exaptation is explored, and the mediating role of boundary-spanning search and the moderating role of absorptive capacity are analyzed.

Hypotheses

Strategic Prospectivity and Exaptation

As an important feature of a “prescient” organization, strategic prospectivity plays a number of important roles in promoting exaptation. (1) Strategic prospectivity provides strategic change cognition and motivation, which are essential for implementing exaptation. Exaptation aims at applying existing technologies and products to new fields or developing new functions of existing products and technologies. As such, exaptation is essentially market innovation or product innovation. Strategic prospectivity advocates innovation and change, and focuses on overall planning and systematic thinking about the future of the organization. Enterprises with strategic prospectivity have a spirit of adventure and a driving force for change. These firms can break free of traditional cognition and development thinking, break through the constraints imposed by old practices and the path dependence of technological innovation, creatively embed existing products and technologies into new fields, and improve the probability of exaptation. (2) Strategic prospectivity provides new triggering opportunities for realizing exaptation. According to Andriani et al. (2017), the more distant a market is, the higher the probability of discovering the unknown will be, and the more dependent on the context of the original function to motivate exaptation companies will be. Strategically forward-looking companies are often committed to the longevity of the enterprise. These companies tend to carry out environmental scanning across physical boundaries, industry boundaries, and technology boundaries to explore new development opportunities. This is done to provide a new trigger environment for the generation of exaptation, and

improve the possibilities of exaptation. (3) Strategic prospectivity provides technical and management support for exaptation. Strategic prospectivity provides new technology support for exaptation, either improving product performance or changing product functions. This results in internal exaptation and full exaptation (Andriani & Carignani, 2014). Meanwhile, strategic prospectivity will also bring about organizational culture innovation, stimulate employees' creativity, realize employees' cognitive diversity, and give birth to diversified innovation methods (Ganzaroli & Pilotti, 2011). In addition, building an organizational structure that connects internal and external resources, and one which realizes the penetration and linking of internal and external resources and capabilities, can also enhance the possibility of new function development (Andriani et al., 2017). Hence, the following hypothesis is proposed:

H1: Strategic prospectivity has a significant positive impact on exaptation.

Mediating Role of Boundary-Spanning Search

Performance feedback theory provides a theoretical basis for interpreting how and why strategic prospectivity drives boundary-spanning search. Performance feedback refers to the problem-oriented search activity caused by the actual performance being lower than the expected performance (Cyert & March, 1963). Strategic prospectivity focuses on cutting-edge technology, developing new products or developing new markets, including trying to become a market leader, taking risks, and developing new products or new markets (Liu & Chen, 2007). According to the performance feedback theory, in order to be engaged in strategic prospectivity, a “prescient” organization must break through the “capability trap” of enterprise innovation and narrow the gap between actual performance and expected performance. Actively implementing cross-industry search to obtain more potential development opportunities, better organizational design, manufacturing processes and innovation approaches has become an indispensable and important strategic choice for corporate decision-makers (Xiong et al., 2011). In addition, Sidhu et al. (2004) found that, compared with reactive, analytical, and defensive organizations, strategic forward-looking organizations have a wider range of product-market areas. It is these firms that are more inclined to actively search for boundary-spanning knowledge to achieve organizational innovation. Moreover, forward-looking organizations emphasize technological advancement. They vigorously promote strategic technological changes, and maintain a leading position through product innovation, market innovation, and technological innovation, which can effectively stimulate boundary-spanning search. Wei et al. (2015) used case studies to explain the mechanism by which strategic prospectivity affects boundary-spanning search. It is believed that strategic prospectivity will drive enterprises to predict the direction of technology development, develop new products, or open up new markets. These activities stimulate enterprises to more actively obtain knowledge resources through boundary-spanning search. Accordingly, the following hypothesis is proposed:

H2: Strategic prospectivity has a significant positive impact on boundary-spanning search.

Boundary-spanning search has the following positive effects on exaptation: (1) Accumulating diverse knowledge helps enterprises to anticipate new potentially unexpected environments and then promote the exaptation process (Katila & Ahuja, 2002). Boundary-spanning search not only can provide rich heterogeneous and complementary knowledge for organizations to update their knowledge base and increase the diversity of knowledge resources but can also provide the market, technology, and management information that organizations need to adjust their knowledge structure in time. This, in turn, results in knowledge agglomeration in different fields, which helps enterprises' R&D personnel to have a deeper understanding and improve their understanding of the concept of "lack" in technology (Xiong et al., 2011). All of this enhances the possibility of exaptation. (2) An important driving force for the generation of exaptation is the user-driven force (Andriani et al., 2017; Ganzaroli & Pilotti, 2011). Using boundary-spanning search to realize two-way interaction with users, enterprises can accurately predict market demand, find potential demand (or marginal customer groups in emerging markets) and tap into new market opportunities. Thus, by combining the existing knowledge system with potential new fields, the use mode of products can be redefined (Andriani et al., 2017), and the new application of existing technologies can be realized. (3) Through the use of boundary-spanning search, enterprises can capture unfamiliar and complementary technical knowledge from suppliers, competitors, customers and other industries. Thus, enterprises can increase the possibility of exchange, sharing and reorganization of technical knowledge in different fields, both inside and outside the enterprise, and they can promote the deconstruction and reconstruction of existing technology. Boundary-spanning search can also help enterprises to find new combinations of existing knowledge elements or reconfigure and combine those elements into a new and more complex form, causing old elements to have new functions (Mokyr, 2000). (4) Exaptation not only emphasizes the application of existing technologies in new fields but also focuses on using existing technologies to adapt to new environments. This requires breaking away from existing routines and breaking the path dependence of product innovation and application. Boundary-spanning search can overcome the local system and cultural barriers and change an enterprise's way of thinking, cognition, and behavior. In addition, boundary-spanning search can break through the traditional thinking of existing business boundaries and break the rigidity of core competencies and organizational inertia (Hu & Yu, 2017). Finally, boundary-spanning search can tap the potential opportunities of existing value network competition, realize the coupling of existing technology and new environment, and open up new markets. Accordingly, the following hypothesis is proposed:

H3: Boundary-spanning search has a significant positive effect on exaptation.

Combining H2 and H3, this paper further infers that strategic prospectivity may indirectly affect exaptation through boundary-spanning search. Hence, the following hypothesis is proposed:

H4: Boundary-spanning search mediates between strategic prospectivity and exaptation.

Moderating Role of Absorptive Capacity

This paper analyzes the regulation mechanism of potential absorptive capacity and real absorptive capacity on the relationship between boundary-spanning search and exaptation and addressed in the following. (1) Enterprises with strong absorptive capacity can more flexibly identify and filter valuable heterogeneous and complementary resources through boundary-spanning search. Such enterprises can integrate old and new knowledge, and use the existing knowledge system to obtain technical advantages in new markets, which in turn improves the possibility of exaptation. At the same time, companies with high potential absorptive capacity are more able to overcome organizational inertia. They can break through geographical boundaries, span longer physical distances, contact, identify, and digest knowledge fields that are far away and unfamiliar and continuously expand and update diversified knowledge reserves. All of this will provide a variety of choices for the application of existing products and the technologies of enterprises in new fields (Mokyr, 2000). In addition, potential absorptive capacity helps enterprises improve the analogical ability of enterprise R&D personnel, with the help of boundary-spanning search. This means enterprises can realize cross links in different knowledge fields, which helps to promote the process of boundary-spanning search-driven exaptation (Mastrogiorgio & Gilsing, 2016). (2) Realized absorptive capacity focuses on transforming external knowledge into an enterprise's internal knowledge and applying that knowledge to the process of product technology innovation and market innovation. Enterprises with strong realized absorptive capacity are more likely to embed the knowledge acquired through boundary-spanning search into the organization. The acquired knowledge will be transformed into knowledge for enterprise innovation and application and integrated into the organizational knowledge system. This will increase the diversity of knowledge, thereby realizing the recombination of internal and external knowledge and promoting exaptation. Therefore, by exploring distant market opportunities (Andriani et al., 2017), enterprises with strong realized absorptive capacity can efficiently process, integrate and transform external knowledge, amass diversified knowledge reserves and integrate them into product market innovation. When an enterprise's existing technology and products are effectively connected with external situational knowledge, this may induce product function transfer, so as to realize exaptation. In addition, exaptation may be generated from a component that has been decomposed by a technology (Dew et al., 2004). Enterprises with strong realized absorptive capacity can improve the innovation and creativity of R&D personnel and enhance their technical decomposition and modification ability by transforming and applying the new knowledge and technology obtained through boundary-spanning search. This will improve the enterprise's understanding of the internal complex dependencies of technical components, enhance their ability to explore additional functions, and stimulate the generation of exaptation. Accordingly, the following hypotheses are proposed:

H5a: Potential absorptive capacity positively moderates the relationship between boundary-spanning search and exaptation.

H5b: Realized absorptive capacity positively moderates the relationship between boundary-spanning search and exaptation.

According to the deduction process of the previous theoretical assumptions, boundary-spanning search plays a mediating role between strategic prospectivity and exaptation (H4). In addition, strategic prospectivity is positively correlated with boundary-spanning search (H2), and boundary-spanning search is positively correlated with exaptation (H3). Absorptive capacity (potential absorptive capacity and realized absorptive capacity) positively moderates the relationship between boundary-spanning search and exaptation (H5a/H5b). This paper further infers that absorptive capacity will moderate the mediating effect of boundary-spanning search between strategic prospectivity and exaptation, thereby forming a moderated mediation model. Hence, the following hypotheses are proposed:

H6a: Potential absorptive capacity positively moderates the mediating effect of boundary-spanning search between strategic prospectivity and exaptation. When the level of potential absorptive capacity is higher, the mediating effect of boundary-spanning search is greater.

H6b: Realized absorptive capacity positively moderates the mediating effect of boundary-spanning search between strategic prospectivity and exaptation. When the level of realized absorptive capacity is higher, the mediating effect of boundary-spanning search is greater.

To sum up, the theoretical model of this paper is shown in Fig. 1.

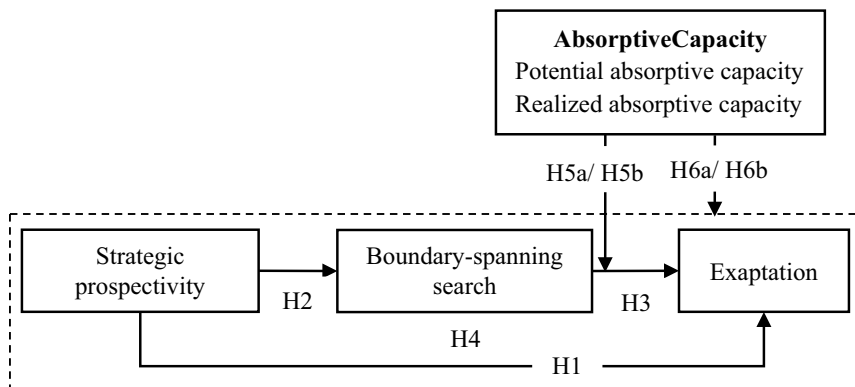


Fig. 1 Theoretical model

Research Design

Sample Data Collection

This paper selected the middle and senior management teams and core business personnel of Chinese manufacturing enterprises in Shanxi Province, Jiangsu Province, and Shanghai as the survey respondents. Specifically, the survey objects include senior managers, middle managers, R&D personnel, and marketing personnel. First, referring to relevant literatures and after consulting experts in the field of innovation and strategic management, a survey questionnaire was prepared. A small-scale pre-test was also conducted to verify that the questionnaire has good reliability and validity. Second, in order to ensure the smooth development of the survey, the surveyed company was contacted in advance. After sufficient communication, agreement was reached on the survey time and method, and the significance of the survey was explained. The authors emphasized that the survey would only be used for academic research, and every effort was made to obtain the support of the investigated enterprises. The survey mainly adopted the methods of on-site answering, on-site recovery, and retention of questionnaires. A total of 350 questionnaires were distributed, and 318 were recovered. After multiple rounds of screening to eliminate invalid questionnaires (according to internal logic, completeness of filling, and other criteria), a total of 276 valid questionnaires were obtained, meaning the effective questionnaire recovery rate was 78.86%. The basic information of the sample is as follows: the industry includes textile and clothing (17.39%), food processing (25.36%), pharmaceutical manufacturing (38.04%), and others (19.20%). The surveyed enterprises have been in operation for 5–15 years (22.83%), 16–30 years (63.04%), and more than 30 years (14.13%). The nature of ownership includes state-owned enterprises (39.13%), private enterprises (12.68%), foreign-funded enterprises (21.38%), collective enterprises (11.23%), and other enterprises (15.58%). The scales of the surveyed enterprises include 300 people or less (8.70%), 300 to 1000 people (13.04%), 1000 to 2000 people (30.43%), 2000 to 3000 people (38.41%), and more than 3000 people (9.42%).

Variable Measurement

The measurement items in this paper mainly come from the mature scales adopted used by studies that made important achievements that have been published in high-quality journals in China and abroad. Alternatively, the measurement items were revised according to relevant literature and were improved by consulting experts in innovation fields, such as enterprises and universities. The measurement indicators are scored using a 5-point Likert scale.

Strategic prospectivity is measured with a four-item scale developed by Liu & Chen (2007). The items include such statements as “Enterprises are sensitive to the signals reflected by the market and take action.” “Enterprises try to become market leaders when launching new products or entering new markets.” “Enterprises are adventurous and always try to launch a large number of products or services

in some target markets with uncertain demand.” “Enterprises always try to develop new products or new markets.” Boundary-spanning is measured with a five-item scale developed by Hu & Yu (2017) and Xu & Yu (2020); the specific items include searches for knowledge from: technical publications and technical conferences, technology licenses, external research institutes, universities or colleges, industry exhibitions, and customers or consumers. Absorptive capacity includes the two categories of potential absorptive capacity and realized absorptive capacity. The scale is adapted from Zahra & George (2002). The three items used to measure potential absorptive capacity are as follows: “Enterprises have a strong ability to identify the value of external knowledge.” “Enterprises introduce external knowledge quickly.” “Enterprises have a strong ability to sort out and classify external knowledge.” The four items used to measure realized absorptive capacity are as follows: “Enterprises have a strong ability to transform external knowledge into their own knowledge.” “Enterprises have a strong ability to apply new knowledge to improve their original technology.” “Enterprises have a strong ability to provide new knowledge to employees, quickly and effectively.” “Enterprises have a strong ability to apply new knowledge to relevant products and services.” Exaptation is measured using a three-item scale developed by Tang et al. (2022). The items are as follows: “A technology of the enterprise has been successfully applied in another field.” “A product of the enterprise has obtained new functions in a new environment.” “The enterprise has new functions through a combination of the functions of existing products.” In addition, drawing on the practice of previous literature (Hu & Yu, 2017; Xu & Yu, 2020), this paper incorporates the control variables firm age, firm size, ownership nature, and industry distribution into the research model.

Reliability, Validity, and Homologous Bias Test

First, regarding reliability, based on the internal consistency of the Cronbach α value test scale, the results reveal that the Cronbach α values of strategic prospectivity, boundary-spanning search, potential absorptive capacity, realized absorptive capacity, and exaptation were 0.736, 0.743, 0.835, 0.708, and 0.760, respectively. All these values are greater than 0.7, clearly indicating that the scale has good internal consistency and high reliability. Second, with regard to validity, discriminant validity is measured by whether or not the square root of the average variation extraction (AVE) of the latent variable is greater than the correlation coefficient between the corresponding latent variables. As shown in Table 1, the square root of the AVE value of each variable is higher than the corresponding correlation coefficient (data in bold diagonal line in Table 1 are the square root values of the AVE value of each variable). Using the AVE value as the test tool of aggregate validity, the AVE values of strategic prospectivity, boundary-spanning search, potential absorptive capacity, realized absorptive capacity, and exaptation were 0.558, 0.502, 0.751, 0.533, and 0.585, respectively. In addition, the AVE values of all variables were greater than 0.50. Therefore, the measurement scale has a good level of validity. In addition, to avoid the problem of homology deviation, the Harman single factor method was used for testing (Hu & Yu, 2017). The results show that, under the condition of no

Table 1 Mean, standard deviation, and correlation coefficient

Variable	1	2	3	4	5	6	7	8	9
Enterprise age	N/A								
Enterprise scale	0.119*	N/A							
Ownership nature	0.015	−0.081	N/A						
Industry distribution	−0.003	0.140**	0.024	N/A					
Strategic prospectivity	0.013	−0.055	0.033	0.010	0.747				
Boundary-spanning search	0.013	0.015	−0.044	−0.090	0.430**	0.709			
Potential absorptive capacity	−0.075	−0.019	−0.029	−0.058	0.234**	0.327**	0.867		
Realized absorptive capacity	0.032	−0.050	0.054	−0.022	0.545**	0.555**	0.262**	0.730	
Exaptation	0.029	−0.060	0.106	−0.026	0.504**	0.471**	0.308**	0.558**	0.765
Mean	3.341	3.565	3.688	3.605	4.073	4.153	3.583	4.177	4.318
Standard deviation	1.112	0.934	0.933	0.961	0.594	0.541	1.069	0.555	0.527

**Significant at the 0.01 level; *significant at the 0.05 level (two-tailed test)

rotation, the first factor only explains 31.403% of the variation. That figure is significantly less than 50% of the statistical requirements, indicating that homology deviation does not affect the data analysis.

Results

Descriptive Statistics and Correlation Analysis of Variables

The results of the descriptive statistics (mean, standard, and deviation) and correlation analysis are shown in Table 2. The correlation coefficient between variables is less than 0.6, and the variance inflation factor is less than 3. These findings indicate that the multicollinearity problem is not significant, and thus, research results are not affected. At the same time, a correlation does exist between the main variables, and this provides a preliminary basis for a test of the theoretical hypotheses.

Mediating Effect Test

The hierarchical linear regression method was used to test the mediation effect, and the detailed results are shown in Table 2. According to Model 2, strategic prospectivity has a significant positive effect on exaptation ($\beta=0.500$, $P<0.001$), and thus, hypothesis 1 is confirmed. According to Model 1, strategic prospectivity has a significant positive impact on boundary-spanning search ($\beta=0.435$, $P<0.001$), and thus, hypothesis 2 is confirmed. According to Model 3, boundary-spanning

Table 2 Regression analysis results

Variable	Exaptation							
	Boundary span- ning search Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Enterprise age	0.002	0.024	0.028	0.023	0.042	0.040	0.015	0.017
Enterprise scale	0.049	−0.024	−0.064	−0.039	−0.062	−0.059	−0.039	−0.035
Ownership nature	−0.052	0.088	0.121*	0.105*	0.123*	0.120*	0.091	0.088
Industry distribution	−0.100	−0.029	0.024	0.003	0.028	0.022	0.009	0.003
Strategic prospectivity	0.435***	0.500***		0.360***				
Boundary-spanning search			0.479***	0.321***	0.421***	0.512***	0.246***	0.332***
Potential absorptive capacity					0.178**	0.135*		
Realized absorptive capacity							0.414***	0.412***
Boundary-spanning search × potential absorptive capacity						0.222***		
Boundary-spanning search × real- ized absorptive capacity								0.245***
ΔR^2	0.199	0.264	0.242	0.347	0.270	0.312	0.359	0.412
F	13.446***	19.410***	17.242***	23.792***	16.585***	17.345***	25.110***	26.805***

***Significant at the 0.001 level; **significant at the 0.01 level; *significant at the 0.05 level

search has a significant positive effect on exaptation ($\beta=0.479$, $P<0.001$), and thus, hypothesis 3 is confirmed. According to Model 4, after adding boundary-spanning search, the regression coefficient of strategic prospectivity on exaptation is significant ($\beta=0.360$, $P<0.001$), and the absolute value of regression coefficient decreased significantly. This finding shows that boundary-spanning search plays a partial intermediary role in the process of strategic prospectivity affecting exaptation, and thus, hypothesis 4 is confirmed.

The hierarchical linear regression method used to test the mediation effect is considered to have the weakest test effect (Hu & Yu, 2017). In this context, Zhao et al. (2010) recommended employing the Bootstrap method introduced by Preacher & Hayes (2004) for testing mediating effects. The Bootstrap method has been widely applied for mediating effect tests in studies published in international academic journals across psychology, organizational behavior, and innovation fields. Therefore, this article will also utilize the Bootstrap method to further assess the effectiveness of the mediating effect of boundary-spanning search. The results show that the mediating effect of boundary-spanning search was 0.149, and the 95% confidence interval (CI) was [0.068, 0.217], excluding 0. These findings indicate that boundary-spanning search has a significant mediating effect. After controlling for the mediating variables, the direct effect of strategic prospectivity on exaptation was 0.416, and the 95% CI was [0.323, 0.510], excluding 0. These findings indicate that the mediating effect is a partial mediator. Combining the hierarchical linear regression method and Bootstrap estimation results, the effectiveness of hypothesis 4 has been further confirmed.

Moderating Effect Test

The hierarchical regression method was used to test the moderating effect of absorptive capacity (Table 2). According to Model 5 and Model 6, potential absorptive capacity positively moderates the relationship between boundary-spanning search and exaptation ($\beta=0.222$, $P<0.001$), and thus, hypothesis 5a is confirmed. According to Model 7 and Model 8, realized absorptive capacity positively moderates the relationship between boundary-spanning search and exaptation ($\beta=0.245$, $P<0.001$), and thus, hypothesis 5b is confirmed. In addition, this paper draws the adjustment effect diagram of absorptive capacity (Figs. 2 and 3). According to Fig. 2, the higher the potential absorptive capacity is, the stronger the impact of boundary-spanning search on exaptation will be. Similarly, according to Figs. 3, the higher the realized absorptive capacity is, the stronger is the impact of boundary-spanning search on exaptation.

Test of Moderated Mediating Effect

According to the bootstrap method proposed by Preacher et al. (2007), the moderated mediation effect test was conducted. The mediating effect of boundary-spanning search between strategic prospectivity and exaptation was analyzed when the level of absorptive capacity was different. Here, the bootstrap method was used for 5000 repeated

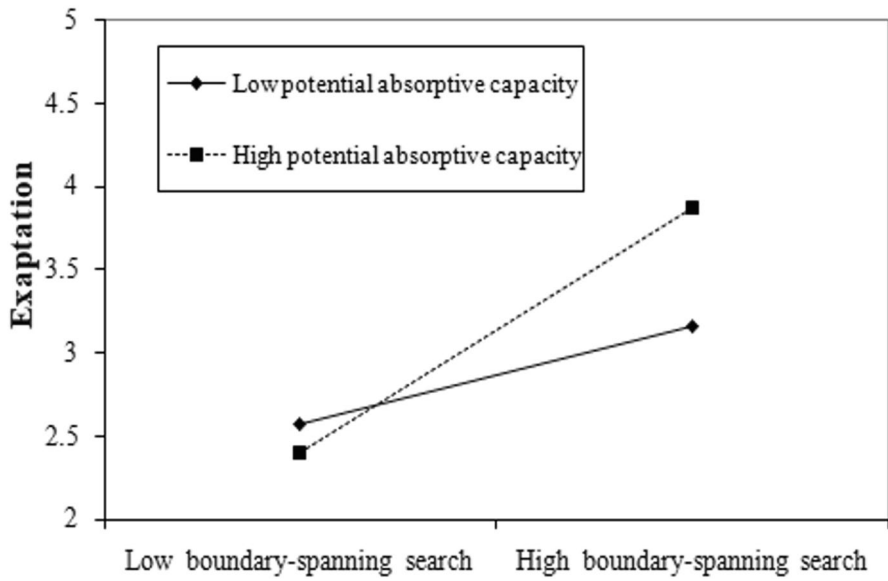


Fig. 2 The moderating effect of potential absorptive capacity on the relationship between boundary-spanning search and exaptation

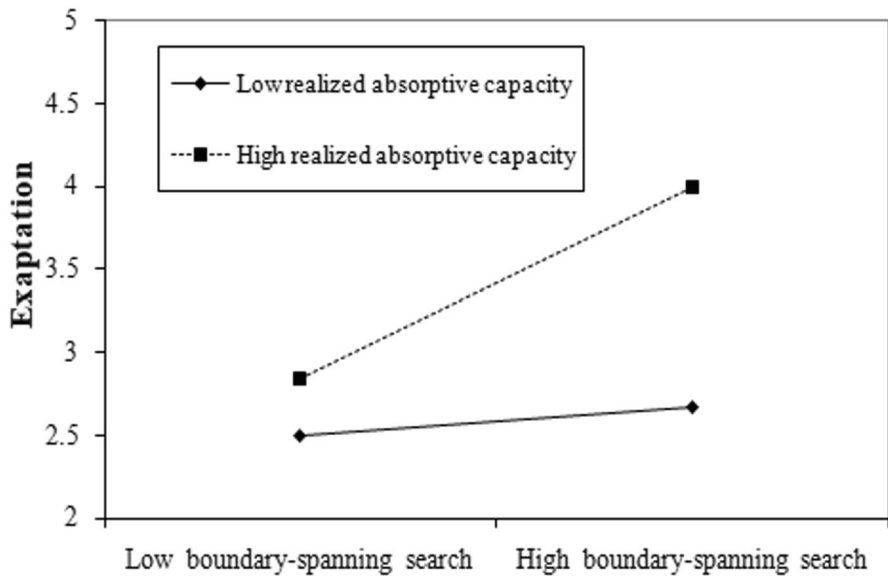


Fig. 3 The moderating effect of realized absorptive capacity on the relationship between boundary-spanning search and exaptation

Table 3 Test results of the regulated mediating effect

Mediation variable	Moderator	Indirect effects under different conditions				Regulated mediating effect			
		Effect	SD	Lower limit set	Upper limit	Index	SD	Lower limit set	Upper limit
Boundary-spanning search	Low-PAC	0.077	0.046	0.024	0.199	0.052	0.026	0.005	0.101
	High-PAC	0.189	0.049	0.104	0.302				
Boundary-spanning search	Low-RAC	0.063	0.050	0.008	0.200	0.077	0.050	-0.036	0.133
	High-RAC	0.148	0.034	0.092	0.232				

PAC potential absorptive capacity, *RAC* realized absorptive capacity

sampling of samples, and the CI was 95%. The test results are shown in Table 3. Under the low potential absorptive capacity level, the mediating effect of boundary-spanning search was established, with a 95% CI [0.024, 0.199]. In addition, at a high potential absorptive capacity level, the mediating effect of boundary-spanning search was established, with a 95% CI [0.104, 0.302]. The test results of the regulated mediating effect demonstrate that its index value was 0.052 (95% CI [0.005, 0.101]), excluding 0. This finding indicates that the moderated intermediary is established. Hypothesis 6a is thus verified, indicating that the indirect effect of strategic prospectivity on exaptation is positively affected by potential absorptive capacity; the stronger the potential absorptive capacity is, the more significant the mediating effect of boundary-spanning search will be. Similarly, at the low level of realized absorptive capacity, the mediating effect of boundary-spanning search was established, with a 95% CI of [0.008, 0.200]. At the high level of realized absorptive capacity, the mediating effect of boundary-spanning search was established, with a 95% CI of [0.092, 0.232]. The test results of the regulated mediating effect demonstrate that its index value was 0.077 (95% CI [-0.036, 0.133]), including 0. This finding suggests that the regulated mediation model was not established. Hypothesis 6b is thus not verified. This shows that the realized absorptive capacity has not significantly adjusted the mediating effect of boundary-spanning search. In addition, the knowledge resources acquired by enterprises through boundary-spanning search can be directly applied within the enterprise, without converting the knowledge.

Discussion and Conclusion

Conclusion

Based on the perspective of boundary-spanning search and absorptive capacity, this paper constructs a theoretical model of the relationship between strategic prospectivity and exaptation. In addition, the role path and boundary conditions of the relationship between the two are explored, and the following conclusions are drawn: Strategic prospectivity positively affects exaptation. The higher the strategic prospectivity is, the more conducive to exaptation. Boundary-spanning search plays a partial mediating role in the relationship between strategic prospectivity and exaptation; strategic prospectivity can also promote exaptation by means of boundary-spanning search. Potential absorptive capacity and realized absorptive capacity positively modulate the relationship between boundary-spanning search and exaptation, respectively, and potential absorptive capacity positively modulates the mediating effect of boundary-spanning search. The hypothesis that realized absorptive capacity positively regulates the intermediary role of boundary-spanning search has not been confirmed needs further testing.

Theoretical Implications

First, identifying strategic prospectivity is an important driving variable for exaptation. Previous literatures have mainly discussed the formation mechanism of exaptation from the perspectives of enterprise mergers and acquisitions (Dew et al., 2004), user innovation (Andriani et al., 2017; Ganzaroli & Pilotti, 2011), knowledge management (Abatecola et al., 2016; Garud et al., 2016), and corporate culture (Ganzaroli et al., 2014). However, the important role of strategic prospectivity on exaptation has not yet been explored. The results show that strategic prospectivity has important value for the generation of exaptation. This is because strategic prospectivity provides strategic change perception, new opportunities and organizational ambidextrous support. In addition, exaptation is an important outcome variable of strategic prospectivity. The research conclusions not only enrich the research results in the field of corporate strategy and exaptation theory; they also provide a new theoretical explanation that gives a more in-depth understanding of the relationship between strategic prospectivity and exaptation in Chinese manufacturing enterprises.

Second, this study explores a new path for strategic prospectivity impact exaptation. Few previous studies have examined the mediating mechanism of boundary-spanning search in the relationship between strategic prospectivity and exaptation. This paper introduces boundary-spanning search between strategic prospectivity and exaptation and examines its mediating effect. The research results reveal the important transmission role of boundary-spanning search in the relationship between strategic prospectivity and exaptation. Strategic prospectivity can promote exaptation through boundary-spanning search. The relevant research conclusions not only echo the conclusions drawn by Sidhu et al. (2004) and Wei et al. (2015); i.e., boundary-spanning search is affected by strategic prospectivity. The conclusions of this study also expand the application scope of boundary-spanning search theory and further reveal the internal logic of strategic prospectivity's effect on exaptation.

Third, the conditions of boundary-spanning search and exaptation are mined in this study. Few previous studies have analyzed the moderating mechanism of absorptive capacity between strategic prospectivity and exaptation. The results of this paper show that absorptive capacity (potential and realized absorptive capacity) has a moderating effect between strategic prospectivity and exaptation. This finding expands the application scenario research of boundary-spanning search.

Last, the boundary conditions of the intermediary mechanism between strategic prospectivity and exaptation are identified. In this paper, potential absorptive capacity and realized absorptive capacity are incorporated into the theoretical framework to analyze and test whether or not absorptive capacity plays a moderate mediating effect in the mechanism of strategic prospectivity on exaptation. The results show that potential absorptive capacity moderates the mediating effect of boundary-spanning search, while realized absorptive capacity does not moderate the mediating role of boundary-spanning search. The former reveals how high potential absorption capacity enhances the degree to which boundary-spanning search is transformed into adaptation. From the perspective of potential absorption capacity, it identifies specific paths from strategic prospectivity and boundary-spanning search to

adaptation. In addition, this conclusion also indicates that the formation of adaptation requires not only forward-looking strategic planning, as well as diverse knowledge links in innovation networks, but also strengthening knowledge management capabilities to coordinate and promote the implementation of adaptation. Moreover, The latter indicates that the realized absorptive capacity has not strengthened the intermediary role of boundary-spanning search and the strategic prospectivity based on the perspective of realized absorptive capacity and the transformation path from boundary-spanning search to adaptation are not established. The possible reason is that the realized absorptive capacity is not sufficient to achieve the conditions for promoting the mediating effect of boundary-spanning search. The research conclusions expand the application scope of absorptive capacity theory and further deepen the situational mechanism research of strategic prospectivity on the mechanism of exaptation.

The findings demonstrate that potential absorptive capacity moderates the mediating effect of boundary-spanning search, whereas realized absorptive capacity does not exert such a moderating influence. The former illustrates how high potential absorptive capacity enhances the extent to which boundary-spanning search transforms into adaptation. Through the lens of potential absorptive capacity, it delineates specific pathways from strategic prospectivity and boundary-spanning search to adaptation. Additionally, this conclusion underscores that the formation of adaptation necessitates not only forward-looking strategic planning and diverse knowledge links within innovation networks, but also strengthened knowledge management capabilities to effectively coordinate and facilitate adaptation implementation.

Conversely, the latter finding indicates that realized absorptive capacity does not bolster the intermediary role of boundary-spanning search. Furthermore, the strategic prospectivity perspective based on realized absorptive capacity and the transformational pathway from boundary-spanning search to adaptation are not firmly established. One plausible explanation is that realized absorptive capacity may not be adequate to facilitate the conditions necessary for enhancing the mediating effect of boundary-spanning search.

Practical Implications

First, strategic prospectivity plays the role of driving exaptation. Senior managers of enterprises should fully recognize the important value of strategic prospectivity. They should study and judge the development trends of the industry, accurately grasp market opportunities, have a long-term vision, make overall planning, design long-term blueprints, and attach importance to new technologies and new markets, in order to win a sustainable competitive advantage. At the same time, the forward-looking characteristics of corporate strategy must be cultivated, and “prescient organizations” must be created. Managers should adhere to a long-term technological innovation strategy, focus on long-distance innovation search across organizational boundaries, and acquire new knowledge, new technologies, new opportunities, and new management ideas and methods. Then, they should form a diversified technology system, establish an organizational structure coordinated with

technological innovation from business, system and management designs within the enterprise, encourage the cultivation of a spirit of risk-taking and change, and be forward-looking and proactive in their enterprises' innovation strategy (Zhang & Lv, 2014), thereby opening a new technological trajectory and industry space.

Second, greater attention should be paid to the value of boundary-spanning search in promoting exaptation. Enterprises should implement boundary-spanning search, improve the ability of external knowledge acquisition, realize the integration and innovation of internal and external knowledge, and enhance their technological innovation abilities. For example, a knowledge management system can be established internally, to manage the heterogeneous and complementary knowledge acquired through boundary-spanning search, improve the efficiency of knowledge integration and utilization, and promote the exaptation process. In addition, enterprises should actively cultivate boundary-spanning search mechanisms, implement forward-looking strategies, and build a platform for knowledge interaction, exchanges and sharing through industry-university-research cooperation. They should also make efforts to realize the coupling and matching of boundary-spanning search capabilities and technological innovation capabilities, and promote exaptation.

Third, enterprises should enhance their absorptive capacity to promote the transformation of boundary-spanning search to exaptation. Attention should be paid to the value of absorptive capacity; organizational absorptive capacity should be cultivated, and knowledge management mechanisms should be improved. Steps should also be taken to create a corporate culture of knowledge acquisition, digestion, and innovative application. In addition, with the help of potential absorptive capacity, information relating to new ideas, new technologies and new markets should be identified and screened. At the same time, full play should be given to the role of realized absorptive capacity, in order to realize the integration, innovation and matching of old and new knowledge, and to promote exaptation.

Limitations and Future Research

Despite the aforementioned contributions to theory and practice, this study still has several potential limitations. First, this paper incorporates boundary-spanning search as a whole concept into the theoretical model. However, the role of boundary-spanning search between strategic prospectivity and exaptation from different dimensions has not been verified. Future research can consider further refining research assumptions from the perspectives of boundary-spanning technology search and boundary-spanning market search. In addition, the differential role of various dimensions of boundary-spanning search in promoting strategic prospectivity and exaptation can be more accurately analyzed. Second, this paper examines the moderating effect of absorptive capacity, and future research can further explore whether there is a regulatory effect from the aspects of environmental dynamics, dynamic ability, and organizational inertia. This approach will further enrich the regulatory mechanism of strategic prospectivity's impact on exaptation.

Acknowledgements Thanks are due to the friends and units who helped to collect the data and to *Journal of the Knowledge Economy* and peer review experts for their opinions.

Author Contribution Conceptualization: CT and YS; methodology: CT; formal analysis and investigation: RS; writing—original draft preparation: CT; writing—review and editing: YS; funding acquisition: CT; resources: CT and YS; supervision: RS.

Funding This study was funded by “Hebei Social Science Fund Project” (Grant number HB21GL035) and “Hebei Agricultural University Introduced Talent Research Project” (Grant number YJ2021002).

Data Availability The raw data supporting the conclusions of this article will be made available by the authors.

Declarations

Ethics Approval Ethical review and approval were not required for the study on human participants in accordance with the local legislation and institutional requirements.

Consent to Participate Written informed consent from the participants was not required to participate in this study in accordance with the national legislation and the institutional requirements.

Conflict of Interest The authors declare no competing interests.

References

- Abatecola, G., Belussi, F., Breslin, D., & Filatotchev, I. (2016). Darwinism, organizational evolution and survival: Key challenges for future research. *Journal of Management & Governance*, 20(1), 1–17. <https://doi.org/10.1007/s10997-015-9310-8>
- Ahuja, G., & Lampert, C. (2001). Entrepreneurship in the large corporation: A longitudinal study of how established firms create breakthrough inventions. *Strategic Management Journal*, 22(6/7), 521–543. <https://doi.org/10.1002/smj.176>
- Andriani, P., Ali, A., & Mastrogiorgio, M. (2017). Measuring exaptation and its impact on innovation, search, and problem solving. *Organization Science*, 28(2), 320–338. <https://doi.org/10.1002/smj.176>
- Andriani, P., & Carignani, G. (2014). Modular exaptation: A missing link in the synthesis of artificial form. *Research Policy*, 43(9), 1608–1620. <https://doi.org/10.1016/j.respol.2014.04.009>
- Arthur, W. B. (2007). The structure of invention. *Research Policy*, 36(2), 274–287. <https://doi.org/10.1016/j.respol.2006.11.005>
- Chen, Z., & Miao, Q. (2014). The impact of external innovation sources on enterprise’s innovation performance: The moderating role of absorptive capacity. *Economic Management*, 36(3), 135–144. <https://doi.org/10.1157/13079921>
- Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1), 128–152. <https://doi.org/10.2307/2393553>
- Cyert, R. M., & March, J. G. (1963). A behavioral theory of the firm. *Journal of Economic Behavior & Organization*, 66(1), 74–94. <https://doi.org/10.1002/bs.3830040202>
- Dew, N., Sarasvathy, S. D., & Venkataraman, S. (2004). The economic implications of exaptation. *Journal of Evolutionary Economics*, 14(1), 69–84. <https://doi.org/10.1007/s00191-003-0180-x>
- Ganzaroli, A., De Noni, I., & Pilotti, L. (2014). The role of social entrepreneurship in leveraging exaptation in locked-in industrial districts: The case of idrogenet in the industrial district of lumezzane. *Innovation the European Journal of Social Science Research*, 27(3), 254–274. <https://doi.org/10.1080/13511610.2014.904742>
- Ganzaroli, A., & Pilotti, L. (2011). Exaptation as source of creativity and innovation[R]. *Departmental Working Papers*.
- Garud, R., Gehman, J., & Giuliani, A. P. (2016). Technological exaptation: A narrative approach. *Industrial and Corporate Change*, 25(1), 149–166. <https://doi.org/10.1093/icc/dtv050>
- Gregory, T. R. (2008). The evolution of complex organs. *Evolution Education & Outreach*, 1(4), 358–389. <https://doi.org/10.1007/s12052-008-0076-1>

- Hu, P., & Yu, B. (2017). Boundary-spanning search, capability reconfiguration and firms' innovation performance—Moderating role of strategic flexibility. *R&D Management*, 29(4), 138–147. <https://doi.org/10.13581/j.cnki.rdm.2017.04.012>
- Kastelli, I., Dimas, P., Stamopoulos, D., & Tsakanikas, A. (2022). Linking digital capacity to innovation performance: The mediating role of absorptive capacity. *Journal of the Knowledge Economy*, 15(1), 238–272. <https://doi.org/10.1007/S13132-022-01092-W>
- Katila, R. A., & Ahuja, G. (2002). Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6), 1183–1194. <https://doi.org/10.2307/3069433>
- Lev, S., Fiegenbaum, A., & Shoham, A. (2009). Managing absorptive capacity stocks to improve performance: Empirical evidence from the turbulent environment of Israeli hospitals. *European Management Journal*, 27(1), 13–25. <https://doi.org/10.1016/j.emj.2008.04.001>
- Liu, H., & Chen, C. (2007). Corporate organizational capital, strategic reform forward-looking and corporate performance: An empirical study based on chinese enterprises. *Management World*, (5), 83–93. <https://doi.org/10.19744/j.cnki.11-1235/f.2007.05.011>
- Mastrogriorgio, M., & Gilsing, V. (2016). Innovation through exaptation and its determinants: The role of technological complexity, analogy making & patent scope. *Research Policy*, 45(7), 1419–1435. <https://doi.org/10.1016/j.respol.2016.04.003>
- Miles, R. E., Snow, C. C., Meyer, A. D., & Jr. (1978). Organizational strategy, structure, and process. *Academy of Management Review*, 3(3), 546–562. <https://doi.org/10.5465/AMR.1978.4305755>
- Mokyr, J. (2000). Evolutionary phenomena in technological change. In J. Ziman (Ed.), *Technological Innovation as an Evolutionary Process* (pp. 52–65). Cambridge University Press.
- Preacher, K. J., Rucker, D. D., & Hayes, A. F. (2007). Addressing moderated mediation hypotheses: Theory, methods, and prescriptions. *Multivariate Behavioral Research*, 42(1), 185–227. <https://doi.org/10.1080/00273170701341316>
- Preacher, K. J., & Hayes, A. F. (2004). SPSS and SAS procedures for estimating indirect effects in simple mediation models. *Behavior Research Methods, Instruments, & Computers*, 36(4), 717–731. <https://doi.org/10.3758/BF03206553>
- Pryor, D., & Henley, A. (2017). Boundary spanning in higher education leadership: Identifying boundaries and practices in a British university. *Studies in Higher Education*, 3, 1–16. <https://doi.org/10.1080/03075079.2017.1318364>
- Ren, S., Gao, T., & Xu, H. (2019). Exaptation as a source of innovation: A literature review. *Foreign Economics & Management*, 41(1), 17–30. <https://doi.org/10.16538/j.cnki.fem.2019.01.002>
- Rothaermel, F. T., & Alexandre, M. T. (2009). Ambidexterity in technology sourcing: The moderating role of absorptive capacity. *Organization Science*, 20(4), 759–780. <https://doi.org/10.1287/orsc.1080.0404>
- Rosenkopf, L., & Nerkar, A. (2001). Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4), 287–306. <https://doi.org/10.1002/smj.160>
- Sidhu, S., Volberda, W., & Commandeur, R. (2004). Exploring exploration orientation and its determinants: Some empirical evidence. *Journal of Management Studies*, 41(6), 913–932. <https://doi.org/10.1111/j.1467-6486.2004.00460.x>
- Tang, C., Shi, Y., & Cai, R. (2022). Correlations of resource bricolage and exaptation with low-cost breakthrough innovations: moderating effect of organizational agility. *Frontiers in Psychology*, 13, 846629.
- Tsai, W. (2001). Knowledge transfer in intraorganizational networks: Effects of network position and absorptive capacity on business unit innovation and performance. *The Academy of Management Journal*, 44(5), 996–1004. <https://doi.org/10.2307/3069443>
- Villani, M., Bonacini, S., Ferrari, D., Serra, R., & Lane, D. (2011). An agent-based model of exaptive processes. *European Management Review*, 4(3), 141–151. <https://doi.org/10.1057/palgrave.emr.1500085>
- Wei, J., Zhang, Y., & Ying, Y. (2015). An evolutionary model among strategic proactiveness, innovative search and innovation performance: Based on a longitudinal case study of sincere group from 1995 to 2012. *Journal of Dialectics of Nature*, 37(4), 88–95. <https://doi.org/10.15994/j.1000-0763.2015.04.013>
- Xiong, W., Feng, X., & Chen, L. (2011). Review and prospect of foreign cross-border search research. *Foreign Economics & Management*, 49(6), 18–26. <https://doi.org/10.16538/j.cnki.fem.2011.06.003>

- Xu, Z., & Yu, G. (2020). A study of the effect of strategic change foresight on the enterprise innovation performance. *Chinese Journal of Management*, 17(8), 1150–1158. <https://doi.org/10.3969/j.issn.1672-884x.2020.08.005>
- Yang, L., & Yu, A. (2016). Empirical study on the relationship between entrepreneurial cognitions and enterprise strategic change prospectivity: The mediating effect of knowledge creation process. *Nankai Business Review*, 19(1), 120–133. <https://doi.org/10.3969/j.issn.1008-3448.2016.01.011>
- Zahra, S. A., & George, G. (2002). Absorptive capacity: A review, reconceptualization, and extension. *Academy of Management Review*, 27(2), 185–203. <https://doi.org/10.2307/4134351>
- Zhang, H., & Lv, S. (2014). Research on relationship of intellectual capital, innovation type and product innovation performance. *Science of Science and Management of S. & T.*, 35(2), 162–168.
- Zhao, X., Jr, J. G. L., & Chen, Q. (2010). Reconsidering baron and kenny: Myths and truths about mediation analysis. *Journal of Consumer Research*, 37(2), 197–206. <https://doi.org/10.1086/651257>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.